

Faculty of Engineering, Chinese University of Hong Kong

# Automated Lesion Segmentation in Whole-Body PET/CT - Multitracer Multicenter Generalization: Datacentrism

Final Report

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# Introduction

Since the end of the last century, PET/CT has become an important tool in diagnosing and managing cancer as well as in treatment planning. Lesion segmentation, an important step for analysing images quantitatively, is done by radiologists and radiation oncologists, making it tedious, expensive and ineffective. Deep Learning, on the other hand, allows us to automate the process of quantitative analysis of PET/CT images such that it can be done quickly. With this in our consideration, we participated in the AutoPET-III challenge held in MICCAI 2024, its task being the automation of Whole-Body PET/CT Lesion Segmentation [1]. Because the size of the dataset is small, making it hard for the model to learn from the data and generalize well, we decided to augment the AutoPET dataset without changing the baseline segmentation model (DynUnet, a MONAI implementation of nnUnet) [2, 3]. To do that, we used a deep generative model to generate new data for data augmentation.

## Related Work

### Towards Generalizable Tumour Synthesis (DiffTumor) [5]

The method proposed here is to use CT images without tumours as well as a lesion segmentation mask to condition the diffusion process to generate CT images with tumours. Using these generated images, they can augment the original tumoured CT images, allowing the model to generalize better.

The whole pipeline consists of 3 steps. In step 1, they build an autoencoder that compresses CT images into a higher-dimension latent space so as to reduce its size and therefore computational power. In this step, we use CT images that are healthy. In step 2, they try to train a latent diffusion model on the latent space to generate CT images with tumours. They condition the diffusion process by CT images without tumours along with a tumour segmentation mask. The tumour segmentation mask is accompanied by an organ segmentation mask here. In Step 3, they generate the CT images with tumours from external datasets of CT images without tumour using the diffusion model in step 2 to perform data augmentation on the original dataset. They then fit the tumour segmentation model on this dataset to boost performance. Using the nnU-Net [3] as the backbone for the segmentation task, their experiment proves the success of this method (Figure 2).

| nnU-Net     | metrics | fold0 | fold1 | fold2 | fold3 | fold4 | average |
|-------------|---------|-------|-------|-------|-------|-------|---------|
| real tumors | DSC (%) | 73.8  | 76.8  | 80.0  | 80.5  | 73.4  | 76.9    |
|             | NSD (%) | 62.7  | 70.2  | 71.2  | 70.8  | 67.5  | 68.5    |
| DiffTumor   | DSC (%) | 84.5  | 83.4  | 81.6  | 83.9  | 77.3  | 82.1    |
|             | NSD (%) | 78.3  | 74.4  | 74.1  | 76.9  | 72.3  | 75.2    |

Figure 2: Tumour segmentation performance of DiffTumor compared to baseline

# Methods

We adapted DiffTumor to our task to conduct data augmentation, then compared the performance of DynUnet trained on the augmented dataset and that of the baseline DynUnet.

## DynUnet baseline

We use the DynUnet [2, 3], which is based on nnU-Net. The nnU-Net fingerprint extractor and planner on the autoPET3 dataset are utilized to configure the model.

## DiffTumor adaptation

The pipeline for data augmentation using a deep generative model borrows largely from DiffTumor [5] but we adapt it to our use. Most notably, the modality changes from CT to PET-CT. In Step 1, whereas DiffTumor trains its autoencoder on a large healthy CT dataset, we train ours only with the AutoPET dataset following the principle of data augmentation, where no new external data should be introduced. In Step 2, unlike DiffTumor which trains only on a certain organ as it wants to demonstrate the generalizability to other organs, we do not need to or intend to demonstrate that. Therefore, Step 2 is trained on the whole Autopet dataset. For Step 2, since we do not have the ground truth organ segmentation mask to accompany the tumour segmentation mask, we utilized SegResNet from “Wholebody ct segmentation” model provided in MONAI model zoo [6-8] to produce a pseudo-organ mask instead. We do not follow Step 3 of DiffTumor since we will use the training settings of DynUnet [3, 9]

## DynUnet adaptation

While almost all other settings are preserved, our training of the DynUnet differs from baseline in that we train not only with real data but also synthesized tumored data.

# Experiments

## Experiment Setting

The AutoPET dataset contains 1614 full-body PET-CT scans, 1291 of which belong to the training set and 323 belong to the validation set. In our experiment, we use the 1291 training data to train the deep generative model for data augmentation. During data augmentation, we generate 3 different synthesized data per 1 training data, thus the augmented training set has a size of 5152 (with some synthesized data from 4 images rejected due to failure of generating pseudo-organ mask by SegResNet).

The input patch size is configured to be (128, 160, 112) with a batch size of 2. Training runs for a total of 581 epochs.

# Results

## Quantitative Results

The table below shows the dice scores for our trained segmentation model (DynUnet) on the baseline dataset and the augmented dataset under different data transformation techniques (and thus different dataset size). We have completed experiments both under the condition of taking 1 random transform per image and 15 random transforms per image.

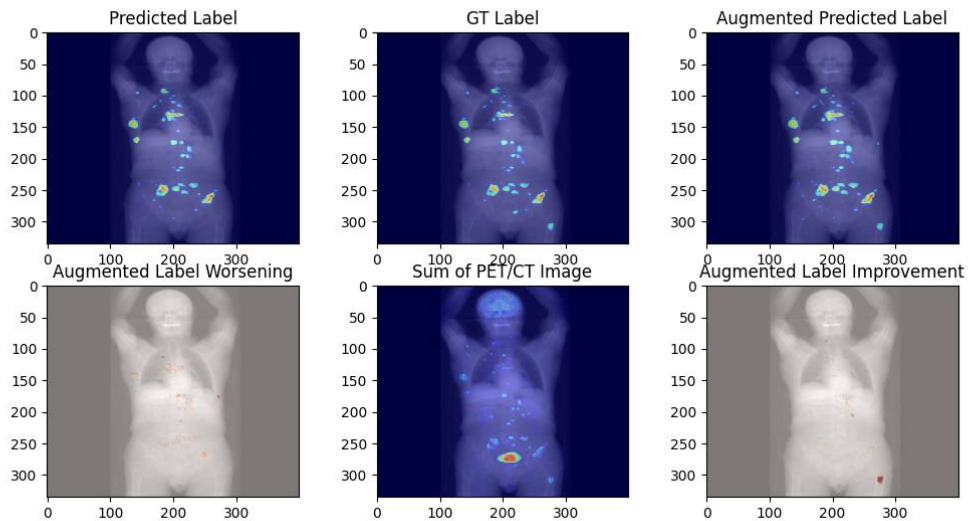
Table 1: Quantitative Comparisons between the Baseline Model and Model trained on the Augmented Dataset

| Baseline/Augmented | Number of transforms per image | Training Dataset size | Dice Score |
|--------------------|--------------------------------|-----------------------|------------|
| Baseline           | 1                              | 1291                  | 0.3650     |
| Augmented          | 1                              | 5152                  | 0.4542     |
| Baseline           | 15                             | 19365                 | 0.5398     |
| Augmented          | 15                             | 77280                 | 0.6143     |

## Qualitative Results

### Predicted Masks

The following are two examples taken from the validation set.



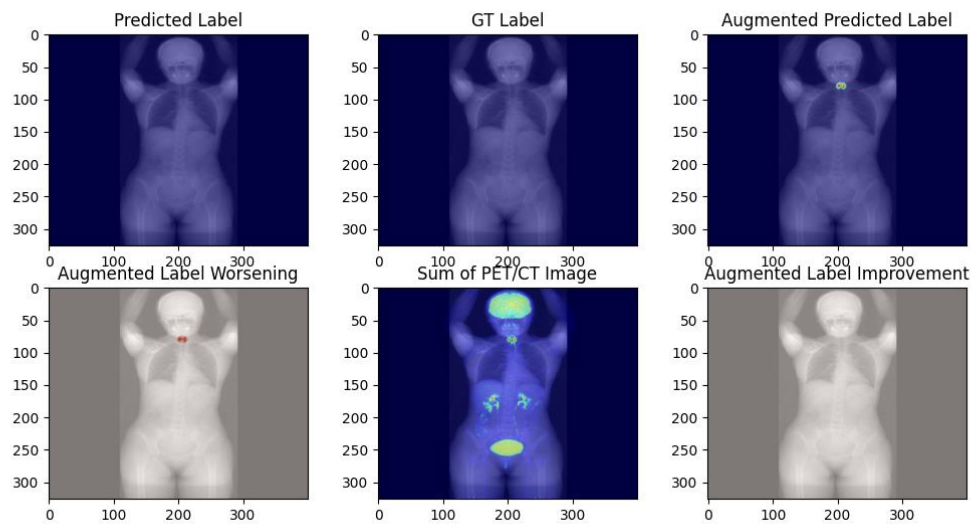
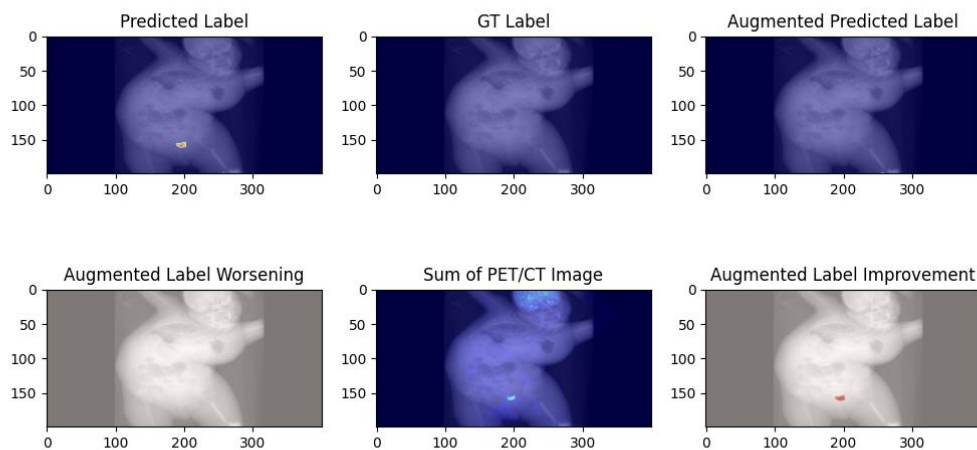
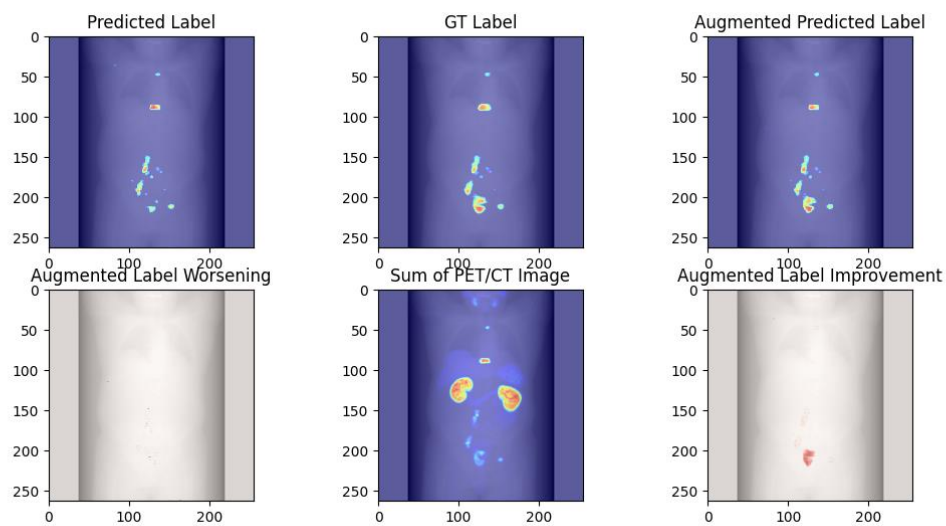
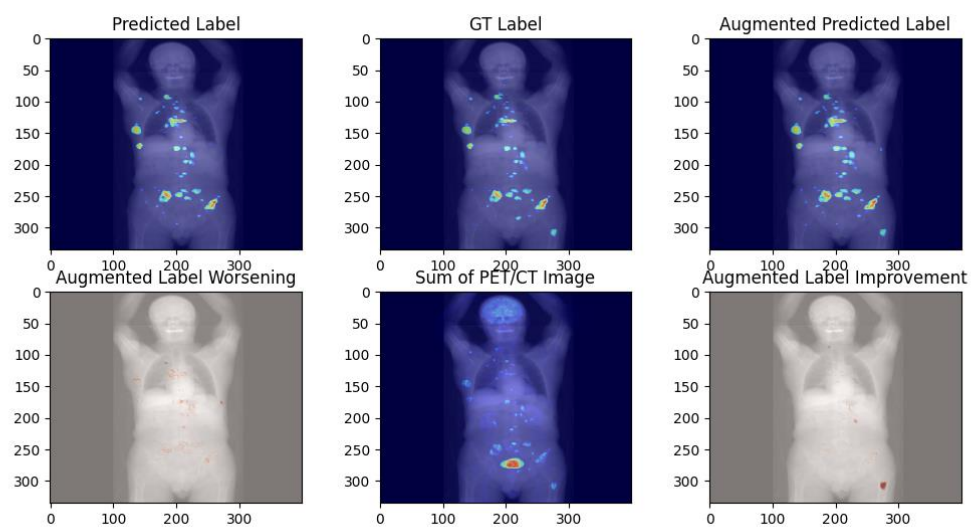
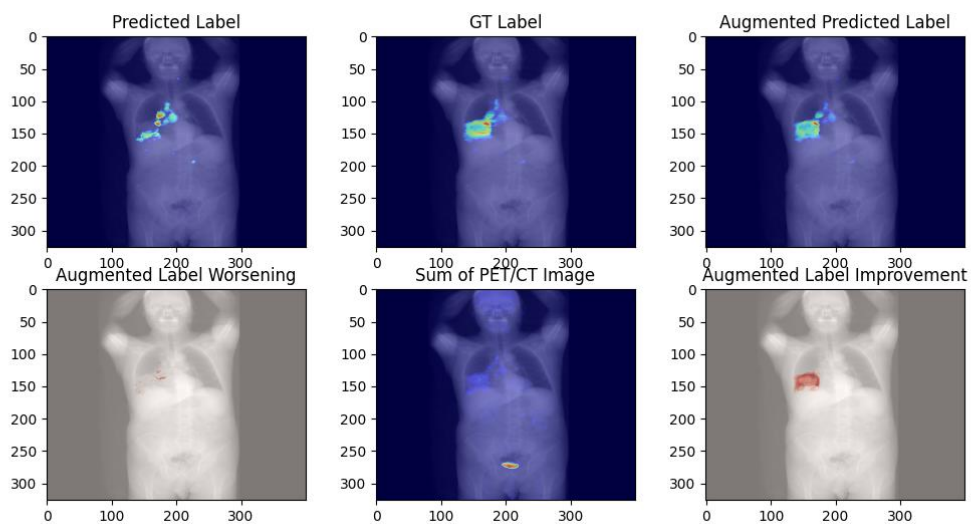


Figure 3: The successful and unsuccessful case of lesion segmentation

The first one is a success in that it improves from the baseline model by successfully segmenting a tumour which the baseline fails to do. The second one, however, presents a false positive displayed by the model trained on the augmented dataset not displayed by the baseline. All in all, while in general, the model trained on the augmented dataset improves from the baseline, there are cases where its performance worsens just as there are cases where it improves. Below are some more examples:





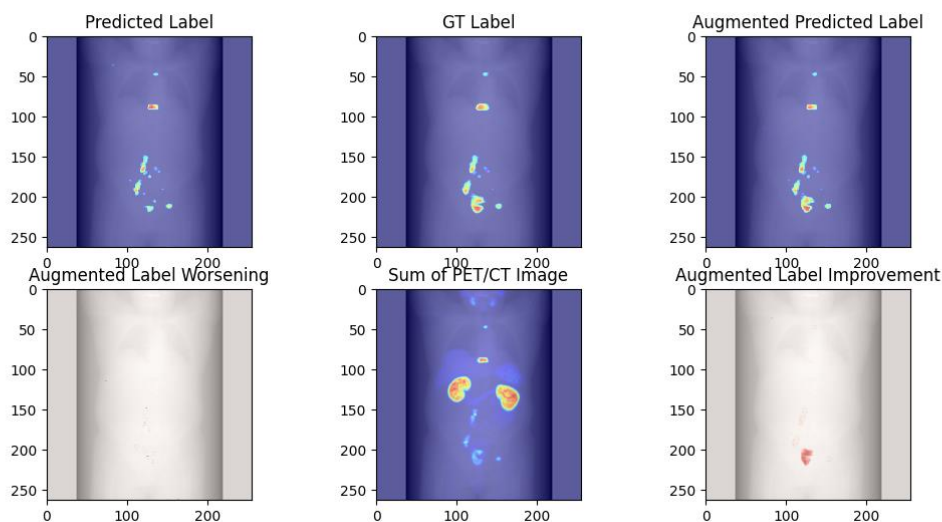


Figure 4: More Lesion Segmentation Results

## Synthesized Tumour

Here are some tumours synthesized on the validation set.

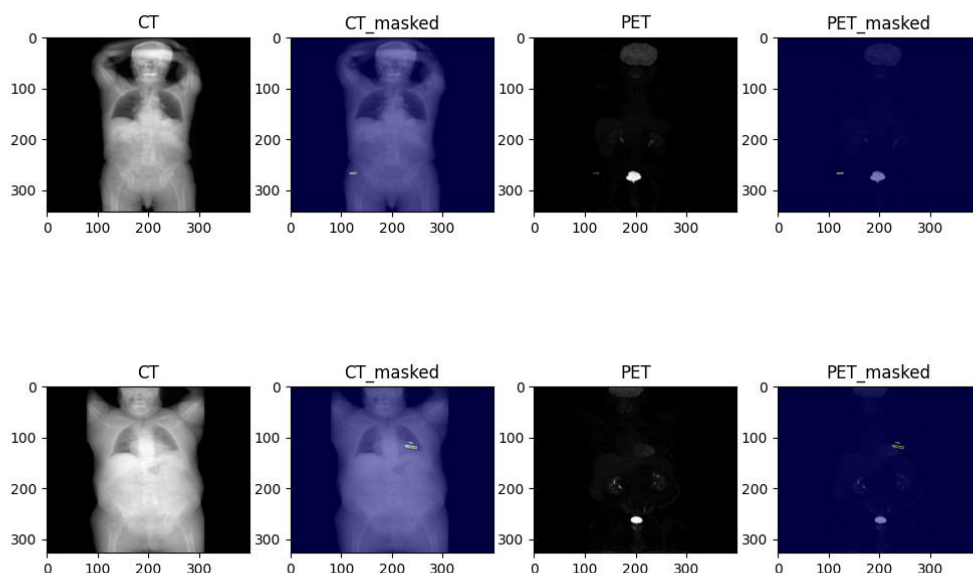


Figure 7: Tumour Generation in unseen data. The generation network learns well by not automatically translating every tumour to be shown as a PET tracer as that is not always the case.

## Conclusion

In conclusion, we demonstrated the feasibility of this new paradigm of data augmentation through using Deep Generative Model to learn the distribution of the existing data and then

synthesize data to augment the original dataset for downstream application in the task of multitracers multicentre full-body PET-CT lesion segmentation. However, further analysis is to be carried out to fully understand our data augmentation method. In our paper, we tested our method of tumour generation with DynUnet backbone (which is based on nnU-Net). We may further test it with other prominent segmentation model backbones such as UNETR [10], Swin-UNETR [11], and even U-KAN [12] to show its generalizability over different backbones.

## References

1. Ingrisch, M., Dexl, J., Jeblick, K., Cyran, C., Gatidis, S., & Kuestner, T.: Automated Lesion Segmentation in Whole-Body PET/CT - Multitracer Multicenter generalization. In: 27th International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI 2024). Zenodo, (Year)
2. Isensee, F., Jäger, P.F., Kohl, S.A., Petersen, J., Maier-Hein, K.H.: Automated design of deep learning methods for biomedical image segmentation. arXiv preprint arXiv:1904.08128 (2019)
3. Isensee, F., Petersen, J., Klein, A., Zimmerer, D., Jaeger, P.F., Kohl, S., Wasserthal, J., Koehler, G., Norajitra, T., Wirkert, S.: nnu-net: Self-adapting framework for u-net-based medical image segmentation. arXiv preprint arXiv:1809.10486 (2018)
4. Wu, L., Zhuang, J., Ni, X., Chen, H.: FreeTumor: Advance Tumor Segmentation via Large-Scale Tumor Synthesis. arXiv preprint arXiv:2406.01264 (2024)
5. Chen, Q., Chen, X., Song, H., Xiong, Z., Yuille, A., Wei, C., Zhou, Z.: Towards generalizable tumor synthesis. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 11147-11158. (Year)
6. Wasserthal, J., Breit, H.-C., Meyer, M.T., Pradella, M., Hinck, D., Sauter, A.W., Heye, T., Boll, D.T., Cyriac, J., Yang, S.: TotalSegmentator: robust segmentation of 104 anatomic structures in CT images. Radiology: Artificial Intelligence 5, (2023)
7. Myronenko, A., Siddiquee, M.M.R., Yang, D., He, Y., Xu, D.: Automated head and neck tumor segmentation from 3D PET/CT HECKTOR 2022 challenge report. 3D Head and Neck Tumor Segmentation in PET/CT Challenge, pp. 31-37. Springer (2022)
8. Tang, Y., Gao, R., Lee, H.H., Han, S., Chen, Y., Gao, D., Nath, V., Bermudez, C., Savona, M.R., Abramson, R.G.: High-resolution 3D abdominal segmentation with random patch network fusion. Medical image analysis 69, 101894 (2021)
9. Myronenko, A.: 3D MRI brain tumor segmentation using autoencoder regularization. In: Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries: 4th International Workshop, BrainLes 2018, Held in Conjunction with MICCAI 2018, Granada, Spain, September 16, 2018, Revised Selected Papers, Part II 4, pp. 311-320. Springer, (Year)
10. Hatamizadeh, A., Tang, Y., Nath, V., Yang, D., Myronenko, A., Landman, B., Roth, H.R., Xu, D.: Unetr: Transformers for 3d medical image segmentation. In: Proceedings of the IEEE/CVF winter conference on applications of computer vision, pp. 574-584. (Year)
11. Tang, Y., Yang, D., Li, W., Roth, H.R., Landman, B., Xu, D., Nath, V., Hatamizadeh, A.: Self-supervised pre-training of swin transformers for 3d medical image analysis. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp. 20730-20740. (Year)
12. Li, C., Liu, X., Li, W., Wang, C., Liu, H., Yuan, Y.: U-KAN Makes Strong Backbone for Medical Image Segmentation and Generation. arXiv preprint arXiv:2406.02918 (2024)



# VeriGuide - Originality Report

## Individual Report

### Background Information

|                          |                                                 |
|--------------------------|-------------------------------------------------|
| Submission Reference ID: | XXXXXXX                                         |
| School / Institution:    | Faculty of Engineering, CUHK                    |
| Year:                    | 2023                                            |
| Term:                    | 2                                               |
| Course:                  | ENGG-0001-A                                     |
| Title:                   | Summer Research Internship 2024                 |
| Assignment Number:       | 2                                               |
| Assignment Marker:       | CHUNG, Katie                                    |
| Student:                 | XXXXXXX                                         |
| Student's School ID:     | XXXXXXX                                         |
| File Name:               | XXXXXXX_Final Report_Rephrased - Veriguide.docx |
| Submitted on:            | 2024-08-14 16:29:15+0800                        |

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| Similarity<br>(at current filter settings): | 4.55%                                                                                                                                                                                                       |
| Current filter settings:                    | Leniency: Default<br>Minimum number of meaningful words in a sentence: 5 (default)<br>Remove Reference Entries (Beta Testing, support English only): Off (default)<br>Some sources are excluded by user: No |
| Sentence(s) Selected By User To 2           |                                                                                                                                                                                                             |
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### Similarity Details

|                     |                                                                                                |
|---------------------|------------------------------------------------------------------------------------------------|
| <b>Index:</b>       | <b>1</b>                                                                                       |
| Suspected Sentence: | To do that, we used a deep generative model to generate new data for data augmentation.        |
| Source Content:     | llama: Improving small language models in domain-specific qa via generative data augmentation. |
| Source:             | <a href="https://arxiv.org/html/2403.02990v1">https://arxiv.org/html/2403.02990v1</a>          |
| From:               | Internet                                                                                       |
| <b>Index:</b>       | <b>2</b>                                                                                       |
| Suspected Sentence: | To do that, we used a deep generative model to generate new data for data augmentation.        |

|                 |                                                                                                 |
|-----------------|-------------------------------------------------------------------------------------------------|
| Source Content: | Tuning language models as training data generators for augmentation-enhanced few-shot learning. |
| Source:         | <a href="https://arxiv.org/html/2403.02990v1">https://arxiv.org/html/2403.02990v1</a>           |
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| <b>Index:</b>       | <b>3</b>                                                                                |
| Suspected Sentence: | To do that, we used a deep generative model to generate new data for data augmentation. |

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| Source Content: | Data augmentation for neural machine translation using generative language model.     |
| Source:         | <a href="https://arxiv.org/html/2403.02990v1">https://arxiv.org/html/2403.02990v1</a> |
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| <b>Index:</b>       | <b>4</b>                                                                                |
| Suspected Sentence: | To do that, we used a deep generative model to generate new data for data augmentation. |

|                 |                                                                                                                       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------|
| Source Content: | (2022b) use T5 as a dialogue user simulator to generate few-shot data augmentations for training conversation models. |
| Source:         | <a href="https://arxiv.org/html/2403.02990v1">https://arxiv.org/html/2403.02990v1</a>                                 |
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| <b>Index:</b>       | <b>5</b>                                                                                                   |
| Suspected Sentence: | In our experiment, we use the 1291 training data to train the deep generative model for data augmentation. |

|                 |                                                                                                 |
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| Source Content: | Tuning language models as training data generators for augmentation-enhanced few-shot learning. |
| Source:         | <a href="https://arxiv.org/html/2403.02990v1">https://arxiv.org/html/2403.02990v1</a>           |
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| <b>Index:</b>       | <b>6</b>                                                                                                   |
| Suspected Sentence: | In our experiment, we use the 1291 training data to train the deep generative model for data augmentation. |

|                 |                                                                                                                                                              |
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| Source Content: | (2022) generates multiple reasoning solutions from LLMs like variants of GPT-3 17B with stochastic sampling to augment the training data for student models. |
| Source:         | <a href="https://arxiv.org/html/2403.02990v1">https://arxiv.org/html/2403.02990v1</a>                                                                        |
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| <b>Index:</b>       | <b>7</b>                                                                                                   |
| Suspected Sentence: | In our experiment, we use the 1291 training data to train the deep generative model for data augmentation. |

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| Source Content: | (2022b) use T5 as a dialogue user simulator to generate few-shot data augmentations for training conversation models. |
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| Login ID                       | XXXXXX                                            |
| Academic Year                  | 2023-2024                                         |
| Term                           | 2                                                 |
| Course Code                    | ENGG-0001-A                                       |
| Course Title                   | Summer Research Internship 2024                   |
| Assignment Marker              | CHUNG, Katie                                      |
| Assignment Number              | 2                                                 |
| Due Date (provided by student) | 2024-08-13                                        |
| Submitted File Name            | XXXXXX_Final Report_Rephrased -<br>Veriguide.docx |
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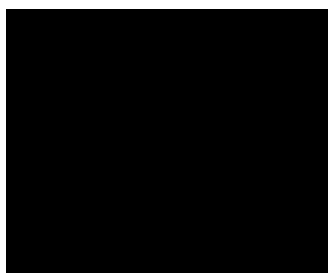
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